

Applications to Random Walk

The key to the developments in the section is that we can embed sums of mean zero independent random variables (see Section 8.1) and martingales (see Section 8.2) into Brownian motion and derive central limit theorems. These results lead in Section 8.3 to CLTs for stationary sequences and in Section 8.4 to the convergence of rescaled empirical distributions to Brownian Bridge. Finally, in Section 8.5 we use the embedding to prove a law of the iterated logarithm.

8.1 Donsker's Theorem

In this section we will prove Donsker's (1951) theorem. The key to its proof is:

Theorem 8.1.1 (Skorokhod's representation theorem) *If $EX = 0$ and $EX^2 < \infty$, then there is a stopping time T for Brownian motion so that $B_T =_d X$ and $ET = EX^2$.*

Remark The Brownian motion in the statement and all the Brownian motions in this section have $B_0 = 0$.

Proof Suppose first that X is supported on $\{a, b\}$, where $a < 0 < b$. Since $EX = 0$, we must have

$$P(X = a) = \frac{b}{b - a} \quad P(X = b) = \frac{-a}{b - a}$$

If we let $T = T_{a,b} = \inf\{t : B_t \notin (a, b)\}$, then Theorem 7.5.3 implies $B_T =_d X$ and Theorem 7.5.5 tells us that

$$ET = -ab = EB_T^2$$

To treat the general case, we will write $F(x) = P(X \leq x)$ as a mixture of two point distributions with mean 0. Let

$$c = \int_{-\infty}^0 (-u) dF(u) = \int_0^{\infty} v dF(v)$$

If φ is bounded and $\varphi(0) = 0$, then using the two formulas for c

$$\begin{aligned}
c \int \varphi(x) dF(x) &= \left(\int_0^\infty \varphi(v) dF(v) \right) \int_{-\infty}^0 (-u) dF(u) \\
&\quad + \left(\int_{-\infty}^0 \varphi(u) dF(u) \right) \int_0^\infty v dF(v) \\
&= \int_0^\infty dF(v) \int_{-\infty}^0 dF(u) (v\varphi(u) - u\varphi(v))
\end{aligned}$$

So we have

$$\int \varphi(x) dF(x) = c^{-1} \int_0^\infty dF(v) \int_{-\infty}^0 dF(u) (v - u) \left\{ \frac{v}{v - u} \varphi(u) + \frac{-u}{v - u} \varphi(v) \right\}$$

The last equation gives the desired mixture. If we let $(U, V) \in \mathbf{R}^2$ have

$$\begin{aligned}
P\{(U, V) = (0, 0)\} &= F(\{0\}) \\
P((U, V) \in A) &= c^{-1} \iint_{(u, v) \in A} dF(u) dF(v) (v - u)
\end{aligned} \tag{8.1.1}$$

for $A \subset (-\infty, 0) \times (0, \infty)$ and define probability measures by $\mu_{0,0}(\{0\}) = 1$ and

$$\mu_{u,v}(\{u\}) = \frac{v}{v - u} \quad \mu_{u,v}(\{v\}) = \frac{-u}{v - u} \quad \text{for } u < 0 < v$$

then

$$\int \varphi(x) dF(x) = E \left(\int \varphi(x) \mu_{U,V}(dx) \right)$$

We proved the last formula when $\varphi(0) = 0$, but it is easy to see that it is true in general. Letting $\varphi \equiv 1$ in the last equation shows that the measure defined in (8.1.1) has total mass 1.

From these calculations it follows that if we have (U, V) with distribution given in (8.1.1) and an independent Brownian motion defined on the same space, then $B(T_{U,V}) =_d X$. Sticklers for detail will notice that $T_{U,V}$ is not a stopping time for B_t , since (U, V) is independent of the Brownian motion. This is not a serious problem, since if we condition on $U = u$ and $V = v$, then $T_{u,v}$ is a stopping time, and this is good enough for all the following calculations. For instance, to compute $E(T_{U,V})$ we observe

$$E(T_{U,V}) = E\{E(T_{U,V} | (U, V))\} = E(-UV)$$

by Theorem 7.5.5, (8.1.1) implies

$$\begin{aligned}
E(-UV) &= \int_{-\infty}^0 dF(u) (-u) \int_0^\infty dF(v) v (v - u) c^{-1} \\
&= \int_{-\infty}^0 dF(u) (-u) \left\{ -u + \int_0^\infty dF(v) c^{-1} v^2 \right\}
\end{aligned}$$

since

$$c = \int_0^\infty v dF(v) = \int_{-\infty}^0 (-u) dF(u)$$

Using the second expression for c now gives

$$E(T_{U,V}) = E(-UV) = \int_{-\infty}^0 u^2 dF(u) + \int_0^{\infty} v^2 dF(v) = EX^2 \quad \square$$

From Theorem 8.1.1, it is only a small step to:

Theorem 8.1.2 *Let $X_1, X_2, \dots$ be i.i.d. with a distribution F , which has mean 0 and variance 1, and let $S_n = X_1 + \dots + X_n$. There is a sequence of stopping times $T_0 = 0, T_1, T_2, \dots$ such that $S_n =_d B(T_n)$ and $T_n - T_{n-1}$ are independent and identically distributed.*

Proof Let $(U_1, V_1), (U_2, V_2), \dots$ be i.i.d. and have distribution given in (8.1.1) and let B_t be an independent Brownian motion. Let $T_0 = 0$, and for $n \geq 1$, let

$$T_n = \inf\{t \geq T_{n-1} : B_t - B(T_{n-1}) \notin (U_n, V_n)\} \quad \square$$

As a corollary of Theorem 8.1.2, we get:

Theorem 8.1.3 (Central limit theorem) *Under the hypotheses of Theorem 8.1.2, $S_n/\sqrt{n} \Rightarrow \chi$, where χ has the standard normal distribution.*

Proof If we let $W_n(t) = B(nt)/\sqrt{n} =_d B_t$ by Brownian scaling, then

$$S_n/\sqrt{n} \stackrel{d}{=} B(T_n)/\sqrt{n} = W_n(T_n/n)$$

The weak law of large numbers implies that $T_n/n \rightarrow 1$ in probability. It should be clear from this that $S_n/\sqrt{n} \Rightarrow B_1$. To fill in the details, let $\epsilon > 0$, pick δ so that

$$P(|B_t - B_1| > \epsilon \text{ for some } t \in (1 - \delta, 1 + \delta)) < \epsilon/2$$

then pick N large enough so that for $n \geq N$, $P(|T_n/n - 1| > \delta) < \epsilon/2$. The last two estimates imply that for $n \geq N$

$$P(|W_n(T_n/n) - W_n(1)| > \epsilon) < \epsilon$$

Since ϵ is arbitrary, it follows that $W_n(T_n/n) - W_n(1) \rightarrow 0$ in probability. Applying the converging together lemma, Lemma 3.2.13, with $X_n = W_n(1)$ and $Z_n = W_n(T_n/n)$, the desired result follows. $\square$

Let $\mathbf{N}$ be the nonnegative integers and let

$$S(u) = \begin{cases} S_k & \text{if } u = k \in \mathbf{N} \\ \text{linear on } [k, k+1] & \text{for } k \in \mathbf{N} \end{cases}$$

Later in this section, we will prove:

Theorem 8.1.4 (Donsker's theorem) *Under the hypotheses of Theorem 8.1.2,*

$$S(n\cdot)/\sqrt{n} \Rightarrow B(\cdot),$$

i.e., the associated measures on $C[0, 1]$ converge weakly.

To motivate ourselves for the proof, we will begin by extracting several corollaries. The key to each one is a consequence of the following result, which follows from Theorem 3.10.1.

Theorem 8.1.5 If $\psi : C[0, 1] \rightarrow \mathbf{R}$ has the property that it is continuous P_0 -a.s., then

$$\psi(S(n\cdot)/\sqrt{n}) \Rightarrow \psi(B(\cdot))$$

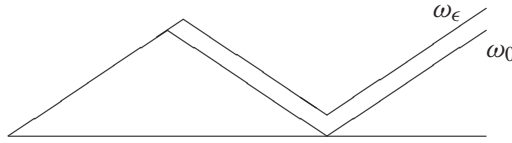
Example 8.1.6 (Maxima) Let $\psi(\omega) = \max\{\omega(t) : 0 \leq t \leq 1\}$. $\psi : C[0, 1] \rightarrow \mathbf{R}$ is continuous, so Theorem 8.1.5 implies

$$\max_{0 \leq m \leq n} S_m/\sqrt{n} \Rightarrow M_1 \equiv \max_{0 \leq t \leq 1} B_t$$

To complete the picture, we observe that by (7.4.4) the distribution of the right-hand side is

$$P_0(M_1 \geq a) = P_0(T_a \leq 1) = 2P_0(B_1 \geq a)$$

Example 8.1.7 (Last 0 before time n) Let $\psi(\omega) = \sup\{t \leq 1 : \omega(t) = 0\}$. This time, ψ is not continuous, for if ω_ϵ with $\omega_\epsilon(0) = 0$ is piecewise linear with slope 1 on $[0, 1/3 + \epsilon]$, -1 on $[1/3 + \epsilon, 2/3]$, and slope 1 on $[2/3, 1]$, then $\psi(\omega_0) = 2/3$ but $\psi(\omega_\epsilon) = 0$ for $\epsilon > 0$.



It is easy to see that if $\psi(\omega) < 1$ and $\omega(t)$ has positive and negative values in each interval $(\psi(\omega) - \delta, \psi(\omega) + \delta)$, then ψ is continuous at ω . By arguments in Subection 8.4.1, the last set has P_0 measure 1. (If the zero at $\psi(\omega)$ was isolated on the left, it would not be isolated on the right.) It follows that

$$\sup\{m \leq n : S_{m-1} \cdot S_m \leq 0\}/n \Rightarrow L = \sup\{t \leq 1 : B_t = 0\}$$

The distribution of L is given in (7.4.7). The last result shows that the arcsine law, Theorem 4.9.5, proved for simple random walks holds when the mean is 0 and variance is finite.

Example 8.1.8 (Occupation times of half-lines) Let

$$\psi(\omega) = |\{t \in [0, 1] : \omega(t) > a\}|.$$

The point $\omega \equiv a$ shows that ψ is not continuous, but it is easy to see that ψ is continuous at paths ω with $|\{t \in [0, 1] : \omega(t) = a\}| = 0$. Fubini's theorem implies that

$$E_0|\{t \in [0, 1] : B_t = a\}| = \int_0^1 P_0(B_t = a) dt = 0$$

so ψ is continuous P_0 -a.s. Using Theorem 8.1.5, we now get that

$$|\{u \leq n : S(u) > a\sqrt{n}\}|/n \Rightarrow |\{t \in [0, 1] : B_t > a\}|$$

As we will now show, with a little work, one can convert this into the more natural result

$$|\{m \leq n : S_m > a\sqrt{n}\}|/n \Rightarrow |\{t \in [0, 1] : B_t > a\}|$$

Proof Application of Theorem 8.1.5 gives that for any a ,

$$|\{t \in [0, 1] : S(nt) > a\sqrt{n}\}| \Rightarrow |\{t \in [0, 1] : B_t > a\}|$$

To convert this into a result about $|\{m \leq n : S_m > a\sqrt{n}\}|$, we note that on $\{\max_{m \leq n} |X_m| \leq \epsilon\sqrt{n}\}$, which by Chebyshev's inequality has a probability $\rightarrow 1$, we have

$$\begin{aligned} |\{t \in [0, 1] : S(nt) > (a + \epsilon)\sqrt{n}\}| &\leq \frac{1}{n} |\{m \leq n : S_m > a\sqrt{n}\}| \\ &\leq |\{t \in [0, 1] : S(nt) > (a - \epsilon)\sqrt{n}\}| \end{aligned}$$

Combining this with the first conclusion of the proof and using the fact that $b \rightarrow |\{t \in [0, 1] : B_t > b\}|$ is continuous at $b = a$ with probability one, one arrives easily at the desired conclusion. $\square$

To compute the distribution of $|\{t \in [0, 1] : B_t > 0\}|$, observe that we proved in Theorem 4.9.6 that if S_n is symmetric simple random walk, then the left-hand side converges to the arcsine law, so the right-hand side has that distribution and is the limit for any random walk with mean 0 and finite variance. The last argument uses an idea called the “invariance principle” that originated with Erdős and Kac (1946, 1947): The asymptotic behavior of functionals of S_n should be the same as long as the central limit theorem applies.

Proof of Theorem 8.1.4 To simplify the proof and prepare for generalizations in the next section, let $X_{n,m}$, $1 \leq m \leq n$, be a triangular array of random variables, $S_{n,m} = X_{n,1} + \dots + X_{n,m}$ and suppose $S_{n,m} = B(\tau_m^n)$. Let

$$S_{n,(u)} = \begin{cases} S_{n,m} & \text{if } u = m \in \{0, 1, \dots, n\} \\ \text{linear for } u \in [m-1, m] & \text{when } m \in \{1, \dots, n\} \end{cases}$$

Lemma 8.1.9 *If $\tau_{[ns]}^n \rightarrow s$ in probability for each $s \in [0, 1]$, then*

$$\|S_{n,(n\cdot)} - B(\cdot)\| \rightarrow 0 \quad \text{in probability}$$

To make the connection with the original problem, let $X_{n,m} = X_m/\sqrt{n}$ and define $\tau_1^n, \dots, \tau_n^n$ so that $(S_{n,1}, \dots, S_{n,n}) =_d (B(\tau_1^n), \dots, B(\tau_n^n))$. If $T_1, T_2, \dots$ are the stopping times defined in the proof of Theorem 8.1.3, Brownian scaling implies $\tau_m^n =_d T_m/n$, so the hypothesis of Lemma 8.1.9 is satisfied.

Proof The fact that B has continuous paths (and hence uniformly continuous on $[0, 1]$) implies that if $\epsilon > 0$, then there is a $\delta > 0$ so that $1/\delta$ is an integer and

$$(a) \quad P(|B_t - B_s| < \epsilon \text{ for all } 0 \leq s \leq 1, |t - s| < 2\delta) > 1 - \epsilon$$

The hypothesis of Lemma 8.1.9 implies that if $n \geq N_\delta$, then

$$P(|\tau_{[nk\delta]}^n - k\delta| < \delta \quad \text{for } k = 1, 2, \dots, 1/\delta) \geq 1 - \epsilon$$

Since $m \rightarrow \tau_m^n$ is increasing, it follows that if $s \in ((k-1)\delta, k\delta)$

$$\begin{aligned} \tau_{[ns]}^n - s &\geq \tau_{[n(k-1)\delta]}^n - k\delta \\ \tau_{[ns]}^n - s &\leq \tau_{[nk\delta]}^n - (k+1)\delta \end{aligned}$$

so if $n \geq N_\delta$,

$$(b) \quad P\left(\sup_{0 \leq s \leq 1} |\tau_{[ns]}^n - s| < 2\delta\right) \geq 1 - \epsilon$$

When the events in (a) and (b) occur

$$(c) \quad |S_{n,m} - B_{m/n}| < \epsilon \text{ for all } m \leq n$$

To deal with $t = (m + \theta)/n$ with $0 < \theta < 1$, we observe that

$$\begin{aligned} |S_{n,(nt)} - B_t| &\leq (1 - \theta)|S_{n,m} - B_{m/n}| + \theta|S_{n,m+1} - B_{(m+1)/n}| \\ &\quad + (1 - \theta)|B_{m/n} - B_t| + \theta|B_{(m+1)/n} - B_t| \end{aligned}$$

Using (c) on the first two terms and (a) on the last two, we see that if $n \geq N_\delta$ and $1/n < 2\delta$, then $\|S_{n,(n\cdot)} - B(\cdot)\| < 2\epsilon$ with probability $\geq 1 - 2\epsilon$. Since ϵ is arbitrary, the proof of Lemma 8.1.9 is complete. $\square$

To get Theorem 8.1.4 now, we have to show:

Lemma 8.1.10 *If φ is bounded and continuous, then $E\varphi(S_{n,(n\cdot)}) \rightarrow E\varphi(B(\cdot))$.*

Proof For fixed $\epsilon > 0$, let $G_\delta = \{\omega : \text{if } \|\omega - \omega'\| < \delta, \text{ then } |\varphi(\omega) - \varphi(\omega')| < \epsilon\}$. Since φ is continuous, $G_\delta \uparrow C[0, 1]$ as $\delta \downarrow 0$. Let $\Delta = \|S_{n,(n\cdot)} - B(\cdot)\|$. The desired result now follows from Lemma 8.1.9 and the trivial inequality

$$|E\varphi(S_{n,(n\cdot)}) - E\varphi(B(\cdot))| \leq \epsilon + (2 \sup |\varphi(\omega)|) \{P(G_\delta^c) + P(\Delta \geq \delta)\} \quad \square$$

To accommodate our final example, we need a trivial generalization of Theorem 8.1.4. Let $C[0, \infty) = \{\text{continuous } \omega : [0, \infty) \rightarrow \mathbf{R}\}$ and let $\mathcal{C}[0, \infty)$ be the σ -field generated by the finite dimensional sets. Given a probability measure μ on $C[0, \infty)$, there is a corresponding measure $\pi_M \mu$ on $C[0, M] = \{\text{continuous } \omega : [0, M] \rightarrow \mathbf{R}\}$ (with $\mathcal{C}[0, M]$ the σ -field generated by the finite dimensional sets) obtained by “cutting off the paths at time M .” Let $(\psi_M \omega)(t) = \omega(t)$ for $t \in [0, M]$ and let $\pi_M \mu = \mu \circ \psi_M^{-1}$. We say that a sequence of probability measures μ_n on $C[0, \infty)$ converges weakly to μ if for all M , $\pi_M \mu_n$ converges weakly to $\pi_M \mu$ on $C[0, M]$, the last concept being defined by a trivial extension of the definitions for $M = 1$. With these definitions, it is easy to conclude:

Theorem 8.1.11 $S(n\cdot)/\sqrt{n} \Rightarrow B(\cdot)$, i.e., the associated measures on $C[0, \infty)$ converge weakly.

Proof By definition, all we have to show is that weak convergence occurs on $C[0, M]$ for all $M < \infty$. The proof of Theorem 8.1.4 works in the same way when 1 is replaced by M . $\square$

Example 8.1.12 Let $N_n = \inf\{m : S_m \geq \sqrt{n}\}$ and $T_1 = \inf\{t : B_t \geq 1\}$. Since $\psi(\omega) = T_1(\omega) \wedge 1$ is continuous P_0 a.s. on $C[0, 1]$ and the distribution of T_1 is continuous, it follows from Theorem 8.1.5 that for $0 < t < 1$

$$P(N_n \leq nt) \rightarrow P(T_1 \leq t)$$

Repeating the last argument with 1 replaced by M and using Theorem 8.1.11 shows that the last conclusion holds for all t .

Exercises

8.1.1 Use Exercise 7.5.4 to conclude that $E(T_{U,V}^2) \leq 4EX^4$.

8.1.2 Suppose S_n is one-dimensional simple random walk and let

$$R_n = 1 + \max_{m \leq n} S_m - \min_{m \leq n} S_m$$

be the number of points visited by time n . Show that $R_n/\sqrt{n} \Rightarrow a$ limit.

8.2 CLTs for Martingales

In this section we will prove central limit theorems for martingales. The key is an embedding theorem due to Strassen (1967). See his Theorem 4.3.

Theorem 8.2.1 *If S_n is a square integrable martingale with $S_0 = 0$, and B_t is a Brownian motion, then there is a sequence of stopping times $0 = T_0 \leq T_1 \leq T_2 \dots$ for the Brownian motion so that*

$$(S_0, S_1, \dots, S_k) \stackrel{d}{=} (B(T_0), B(T_1), \dots, B(T_k)) \quad \text{for all } k \geq 0$$

Proof We include $S_0 = 0 = B(T_0)$ only for the sake of starting the induction argument. Suppose we have $(S_0, \dots, S_{k-1}) \stackrel{d}{=} (B(T_0), \dots, B(T_{k-1}))$ for some $k \geq 1$. The strong Markov property implies that $\{B(T_{k-1} + t) - B(T_{k-1}) : t \geq 0\}$ is a Brownian motion that is independent of $\mathcal{F}(T_{k-1})$. Let $\mu_k(s_0, \dots, s_{k-1}; \cdot)$ be a regular conditional distribution of $S_k - S_{k-1}$, given $S_j = s_j$, $0 \leq j \leq k-1$, that is, for each Borel set A

$$P(S_k - S_{k-1} \in A | S_0, \dots, S_{k-1}) = \mu_k(S_0, \dots, S_{k-1}; A)$$

By Exercises 4.1.18 and 4.1.16, this exists, and we have

$$0 = E(S_k - S_{k-1} | S_0, \dots, S_{k-1}) = \int x \mu_k(S_0, \dots, S_{k-1}; dx)$$

so the mean of the conditional distribution is 0 almost surely. Using Theorem 8.1.1 now, we see that for almost every $\hat{S} \equiv (S_0, \dots, S_{k-1})$, there is a stopping time $\tau_{\hat{S}}$ (for $\{B(T_{k-1} + t) - B(T_{k-1}) : t \geq 0\}$) so that

$$B(T_{k-1} + \tau_{\hat{S}}) - B(T_{k-1}) \stackrel{d}{=} \mu_k(S_0, \dots, S_{k-1}; \cdot)$$

If we let $T_k = T_{k-1} + \tau_{\hat{S}}$, then $(S_0, \dots, S_k) \stackrel{d}{=} (B(T_0), \dots, B(T_k))$ and the result follows by induction. $\square$

Remark While the details of the proof are fresh in the reader's mind, we would like to observe that if $E(S_k - S_{k-1})^2 < \infty$, then

$$E(\tau_{\hat{S}} | S_0, \dots, S_{k-1}) = \int x^2 \mu_k(S_0, \dots, S_{k-1}; dx)$$

since $B_t^2 - t$ is a martingale and $\tau_{\hat{S}}$ is the exit time from a randomly chosen interval $(S_{k-1} + U_k, S_{k-1} + V_k)$.

Before getting involved in the details of the central limit theorem, we show that the existence of the embedding allows us to give an easy proof of Theorem 4.5.2.

Theorem 8.2.2 Let $\mathcal{F}_m = \sigma(S_0, S_1, \dots, S_m)$. $\lim_{n \rightarrow \infty} S_n$ exists and is finite on $\sum_{m=1}^{\infty} E((S_m - S_{m-1})^2 | \mathcal{F}_{m-1}) < \infty$.

Proof Let $\mathcal{B}_t$ be the filtration generated by Brownian motion, and let $t_m = T_m - T_{m-1}$. By construction we have

$$E((S_m - S_{m-1})^2 | \mathcal{F}_{m-1}) = E(t_m | \mathcal{B}(T_{m-1}))$$

Let $M = \inf\{n : \sum_{m=1}^{n+1} E(t_m | \mathcal{B}(T_{m-1})) > A\}$. M is a stopping time and by its definition has $\sum_{m=1}^M E(t_m | \mathcal{B}(T_{m-1})) \leq A$. Now $\{M \geq m\} \in \mathcal{F}_{m-1}$, so

$$\begin{aligned} E \sum_{m=1}^M t_m &= \sum_{m=1}^{\infty} E(t_m; M \geq m) \\ &= \sum_{m=1}^{\infty} E(E(t_m | \mathcal{B}(T_{m-1}); M \geq m)) = E \sum_{m=1}^M E(t_m | \mathcal{B}(T_{m-1})) \leq A \end{aligned}$$

From this we see that $\sum_{m=1}^M t_m < \infty$. As $A \rightarrow \infty$, $P(M < \infty) \downarrow 0$, so $T_{\infty} = \sum_{m=1}^{\infty} t_m < \infty$. Since $B(T_n) \rightarrow B(T_{\infty})$ as $n \rightarrow \infty$ the desired result follows. $\square$

Our first step toward the promised central limit theorem is to prove a result of Freedman (1971a). We say that $X_{n,m}$, $\mathcal{F}_{n,m}$, $1 \leq m \leq n$, is a **martingale difference array** if $X_{n,m} \in \mathcal{F}_{n,m}$ and $E(X_{n,m} | \mathcal{F}_{n,m-1}) = 0$ for $1 \leq m \leq n$, where $\mathcal{F}_{n,0} = \{\emptyset, \Omega\}$. Let $S_{n,m} = X_{n,1} + \dots + X_{n,m}$. Let $N = \{0, 1, 2, \dots\}$. Throughout the section, $S_{n,(\cdot)}$ denotes the linear interpolation of $S_{n,m}$ defined by

$$S_{n,(u)} = \begin{cases} S_k & \text{if } u = k \in N \\ \text{linear for } u \in [k, k+1] & \text{when } k \in N \end{cases}$$

and $B(\cdot)$ is a Brownian motion with $B_0 = 0$. Let

$$V_{n,k} = \sum_{1 \leq m \leq k} E(X_{n,m}^2 | \mathcal{F}_{n,m-1})$$

Theorem 8.2.3 Suppose $\{X_{n,m}, \mathcal{F}_{n,m}\}$ is a martingale difference array.

If (i) for each t , $V_{n,[nt]} \rightarrow t$ in probability,

(i) $|X_{n,m}| \leq \epsilon_n$ for all m with $\epsilon_n \rightarrow 0$, and

then $S_{n,(\cdot)} \Rightarrow B(\cdot)$.

Proof (i) implies $V_{n,n} \rightarrow 1$ in probability. By stopping each sequence at the first time $V_{n,k} > 2$ and setting the later $X_{n,m} = 0$, we can suppose without loss of generality that $V_{n,n} \leq 2 + \epsilon_n^2$ for all n . By Theorem 8.2.1, we can find stopping times $T_{n,1}, \dots, T_{n,n}$ so that $(S_{n,1}, \dots, S_{n,n}) =_d (B(T_{n,1}), \dots, B(T_{n,n}))$. By Lemma 8.1.9, it suffices to show that $T_{n,[nt]} \rightarrow t$ in probability for each $t \in [0, 1]$. To do this we let $t_{n,m} = T_{n,m} - T_{n,m-1}$ (with

$T_{n,0} = 0$) observe that by the remark after the proof of Theorem 8.2.1, $E(t_{n,m}|\mathcal{F}_{n,m-1}) = E(X_{n,m}^2|\mathcal{F}_{n,m-1})$. The last observation and hypothesis (i) imply

$$\sum_{m=1}^{[nt]} E(t_{n,m}|\mathcal{F}_{n,m-1}) \rightarrow t \quad \text{in probability}$$

To get from this to $T_{n,[nt]} \rightarrow t$ in probability, we observe

$$E \left(\sum_{m=1}^{[nt]} t_{n,m} - E(t_{n,m}|\mathcal{F}_{n,m-1}) \right)^2 = E \sum_{m=1}^{[nt]} \{t_{n,m} - E(t_{n,m}|\mathcal{F}_{n,m-1})\}^2$$

by the orthogonality of martingale increments, Theorem 4.4.7. Now

$$\begin{aligned} E(\{t_{n,m} - E(t_{n,m}|\mathcal{F}_{n,m-1})\}^2|\mathcal{F}_{n,m-1}) &\leq E(t_{n,m}^2|\mathcal{F}_{n,m-1}) \\ &\leq C E(X_{n,m}^4|\mathcal{F}_{n,m-1}) \leq C \epsilon_n^2 E(X_{n,m}^2|\mathcal{F}_{n,m-1}) \end{aligned}$$

by Exercise 7.5.4 and assumption (ii). Summing over n , taking expected values, and recalling we have assumed $V_{n,n} \leq 2 + \epsilon_n^2$, it follows that

$$E \left(\sum_{m=1}^{[nt]} t_{n,m} - E(t_{n,m}|\mathcal{F}_{n,m-1}) \right)^2 \rightarrow C \epsilon_n^2 E V_{n,n} \rightarrow 0$$

Unscrambling the definitions, we have shown $E(T_{n,[nt]} - V_{n,[nt]})^2 \rightarrow 0$, so Chebyshev's inequality implies $P(|T_{n,[nt]} - V_{n,[nt]}| > \epsilon) \rightarrow 0$, and using (ii) now completes the proof. $\square$

With Theorem 8.2.3 established, a truncation argument gives us:

Theorem 8.2.4 (Lindeberg-Feller theorem for martingales) Suppose $X_{n,m}$, $\mathcal{F}_{n,m}$, $1 \leq m \leq n$ is a martingale difference array.

If (i) $V_{n,[nt]} \rightarrow t$ in probability for all $t \in [0, 1]$ and

(ii) for all $\epsilon > 0$, $\sum_{m \leq n} E(X_{n,m}^2 1_{(|X_{n,m}| > \epsilon)}|\mathcal{F}_{n,m-1}) \rightarrow 0$ in probability,

then $S_{n,(n \cdot)} \Rightarrow B(\cdot)$.

Remark 8.2.1 If $X_{n,m}$ on each row are independent, this reduces to the martingale central limit theorem.

Proof The first step is to truncate so that we can apply Theorem 8.2.3. Let

$$\hat{V}_n(\epsilon) = \sum_{m=1}^n E(X_{n,m}^2 1_{(|X_{n,m}| > \epsilon_n)}|\mathcal{F}_{n,m-1})$$

Lemma 8.2.5 If $\epsilon_n \rightarrow 0$ slowly enough, then $\epsilon_n^{-2} \hat{V}_n(\epsilon_n) \rightarrow 0$ in probability.

Remark The ϵ_n^{-2} in front is so that we can conclude

$$\sum_{m \leq n} P(|X_{n,m}| > \epsilon_n|\mathcal{F}_{n,m-1}) \rightarrow 0 \text{ in probability}$$

Proof Let N_m be chosen so that $P(m^2 \hat{V}_n(1/m) > 1/m) \leq 1/m$ for $n \geq N_m$. Let $\epsilon_n = 1/m$ for $n \in [N_m, N_{m+1})$ and $\epsilon_n = 1$ if $n < N_1$. If $\delta > 0$ and $1/m < \delta$, then for $n \in [N_m, N_{m+1})$

$$P(\epsilon_n^{-2} \hat{V}_n(\epsilon_n) > \delta) \leq P(m^2 \hat{V}_n(1/m) > 1/m) \leq 1/m \quad \square$$

Let $\hat{X}_{n,m} = X_{n,m} 1_{(|X_{n,m}| > \epsilon_n)}$, $\bar{X}_{n,m} = X_{n,m} 1_{(|X_{n,m}| \leq \epsilon_n)}$ and

$$\tilde{X}_{n,m} = \bar{X}_{n,m} - E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})$$

Our next step is to show:

Lemma 8.2.6 *If we define $\tilde{S}_{n,(n\cdot)}$ in the obvious way, then Theorem 8.2.3 implies $\tilde{S}_{n,(n\cdot)} \Rightarrow B(\cdot)$.*

Proof Since $|\tilde{X}_{n,m}| \leq 2\epsilon_n$, we only have to check (ii) in Theorem 8.2.3. To do this, we observe that the conditional variance formula, Theorem 4.4.8, implies

$$E(\tilde{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) = E(\bar{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) - E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})^2$$

For the first term, we observe

$$E(\bar{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) = E(X_{n,m}^2 | \mathcal{F}_{n,m-1}) - E(\hat{X}_{n,m}^2 | \mathcal{F}_{n,m-1})$$

For the second, we observe that $E(X_{n,m} | \mathcal{F}_{n,m-1}) = 0$ implies

$$E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})^2 = E(\hat{X}_{n,m} | \mathcal{F}_{n,m-1})^2 \leq E(\hat{X}_{n,m}^2 | \mathcal{F}_{n,m-1})$$

by Jensen's inequality, so it follows from (a) and (i) that

$$\sum_{m=1}^{[nt]} E(\tilde{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) \rightarrow t \quad \text{for all } t \in [0, 1] \quad \square$$

Having proved Lemma 8.2.6, it remains to estimate the difference between $S_{n,(n\cdot)}$ and $\tilde{S}_{n,(n\cdot)}$. On $\{|X_{n,m}| \leq \epsilon_n \text{ for all } 1 \leq m \leq n\}$, we have

$$\|S_{n,(n\cdot)} - \tilde{S}_{n,(n\cdot)}\| \leq \sum_{m=1}^n |E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})| \quad (8.2.1)$$

To handle the right-hand side, we observe

$$\begin{aligned} \sum_{m=1}^n |E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})| &= \sum_{m=1}^n |E(\hat{X}_{n,m} | \mathcal{F}_{n,m-1})| \\ &\leq \sum_{m=1}^n E(|\hat{X}_{n,m}| | \mathcal{F}_{n,m-1}) \\ &\leq \epsilon_n^{-1} \sum_{m=1}^n E(\hat{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) \rightarrow 0 \end{aligned} \quad (8.2.2)$$

in probability by Lemma 8.2.5. To complete the proof now, it suffices to show

$$P(|X_{n,m}| > \epsilon_n \text{ for some } m \leq n) \rightarrow 0 \quad (8.2.3)$$

for with (8.2.2) and (8.2.1) this implies $\|S_{n,(n\cdot)} - \tilde{S}_{n,(n\cdot)}\| \rightarrow 0$ in probability. The proof of Theorem 8.2.3 constructs a Brownian motion with $\|\tilde{S}_{n,(n\cdot)} - B(\cdot)\| \rightarrow 0$, so the desired result follows from the triangle inequality and Lemma 8.1.10.

To prove (8.2.3), we will use Lemma 3.5 of Dvoretzky (1972).

Lemma 8.2.7 *If A_n is adapted to $\mathcal{G}_n$, then for any nonnegative $\delta \in \mathcal{G}_0$,*

$$P\left(\bigcup_{m=1}^n A_m | \mathcal{G}_0\right) \leq \delta + P\left(\sum_{m=1}^n P(A_m | \mathcal{G}_{m-1}) > \delta \mid \mathcal{G}_0\right)$$

Proof We proceed by induction. When $n = 1$, the conclusion says

$$P(A_1 | \mathcal{G}_0) \leq \delta + P(P(A_1 | \mathcal{G}_0) > \delta | \mathcal{G}_0)$$

This is obviously true on $\Omega_- \equiv \{P(A_1 | \mathcal{G}_0) \leq \delta\}$ and also on $\Omega_+ \equiv \{P(A_1 | \mathcal{G}_0) > \delta\} \in \mathcal{G}_0$, since on Ω_+

$$P(P(A_1 | \mathcal{G}_0) > \delta | \mathcal{G}_0) = 1 \geq P(A_1 | \mathcal{G}_0)$$

To prove the result for n sets, observe that by the last argument the inequality is trivial on Ω_+ . Let $B_m = A_m \cap \Omega_-$. Since $\Omega_- \in \mathcal{G}_0 \subset \mathcal{G}_{m-1}$, $P(B_m | \mathcal{G}_{m-1}) = P(A_m | \mathcal{G}_{m-1})$ on Ω_- . (See Exercise 4.1.2.) Applying the result for $n - 1$ sets with $\gamma = \delta - P(B_1 | \mathcal{G}_0) \geq 0$,

$$P\left(\bigcup_{m=2}^n B_m | \mathcal{G}_1\right) \leq \gamma + P\left(\sum_{m=2}^n P(B_m | \mathcal{G}_{m-1}) > \gamma \mid \mathcal{G}_1\right)$$

Taking conditional expectation w.r.t. $\mathcal{G}_0$ and noting $\gamma \in \mathcal{G}_0$,

$$P\left(\bigcup_{m=2}^n B_m | \mathcal{G}_0\right) \leq \gamma + P\left(\sum_{m=1}^n P(B_m | \mathcal{G}_{m-1}) > \delta \mid \mathcal{G}_0\right)$$

$\bigcup_{2 \leq m \leq n} B_m = (\bigcup_{2 \leq m \leq n} A_m) \cap \Omega_-$ and another use of Exercise 4.1.2 shows

$$\sum_{1 \leq m \leq n} P(B_m | \mathcal{G}_{m-1}) = \sum_{1 \leq m \leq n} P(A_m | \mathcal{G}_{m-1})$$

on Ω_- . So, on Ω_- ,

$$P\left(\bigcup_{m=2}^n A_m | \mathcal{G}_0\right) \leq \delta - P(A_1 | \mathcal{G}_0) + P\left(\sum_{m=1}^n P(A_m | \mathcal{G}_{m-1}) > \delta | \mathcal{G}_0\right)$$

The result now follows from

$$P\left(\bigcup_{m=1}^n A_m | \mathcal{G}_0\right) \leq P(A_1 | \mathcal{G}_0) + P\left(\bigcup_{m=2}^n A_m | \mathcal{G}_0\right)$$

To see this, let $C = \bigcup_{2 \leq m \leq n} A_m$, observe that $1_{A_1 \cup C} \leq 1_{A_1} + 1_C$, and use the monotonicity of conditional expectations. $\square$

Proof of (8.2.3) Let $A_m = \{|X_{n,m}| > \epsilon_n\}$, $\mathcal{G}_m = \mathcal{F}_{n,m}$, and let δ be a positive number. Lemma 8.2.7 implies

$$P(|X_{n,m}| > \epsilon_n \text{ for some } m \leq n) \leq \delta + P\left(\sum_{m=1}^n P(|X_{n,m}| > \epsilon_n | \mathcal{F}_{n,m-1}) > \delta\right)$$

To estimate the right-hand side, we observe that “Chebyshev’s inequality” (Exercise 1.3 in Chapter 4) implies

$$\sum_{m=1}^n P(|X_{n,m}| > \epsilon_n | \mathcal{F}_{n,m-1}) \leq \epsilon_n^{-2} \sum_{m=1}^n E(\hat{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) \rightarrow 0$$

so $\limsup_{n \rightarrow \infty} P(|X_{n,m}| > \epsilon_n \text{ for some } m \leq n) \leq \delta$. Since δ is arbitrary, the proof of (e) and hence of Theorem 8.2.4 is complete. $\square$

For applications, it is useful to have a result for a single sequence.

Theorem 8.2.8 (Martingale central limit theorem) Suppose $X_n, \mathcal{F}_n, n \geq 1$, is a martingale difference sequence and let $V_k = \sum_{1 \leq n \leq k} E(X_n^2 | \mathcal{F}_{n-1})$.

If (i) $V_k/k \rightarrow \sigma^2 > 0$ in probability and

(ii) $n^{-1} \sum_{m \leq n} E(X_m^2 1_{(|X_m| > \epsilon \sqrt{n})}) \rightarrow 0$

then $S_{(n)}/\sqrt{n} \Rightarrow \sigma B(\cdot)$.

Proof Let $X_{n,m} = X_m/\sigma\sqrt{n}$, $\mathcal{F}_{n,m} = \mathcal{F}_m$. Changing notation and letting $k = nt$, our first assumption becomes (i) of Theorem 8.2.4. To check (ii), observe that

$$E \sum_{m=1}^n E(X_{n,m}^2 1_{(|X_{n,m}| > \epsilon) | \mathcal{F}_{n,m-1}}) = \sigma^{-2} n^{-1} \sum_{m=1}^n E(X_m^2 1_{(|X_m| > \epsilon \sigma \sqrt{n})}) \rightarrow 0 \quad \square$$

8.3 CLTs for Stationary Sequences

We begin by considering the martingale case.

Theorem 8.3.1 Suppose $X_n, n \in \mathbf{Z}$, is an ergodic stationary sequence of square integrable martingale differences, i.e., $\sigma^2 = EX_n^2 < \infty$ and $E(X_n | \mathcal{F}_{n-1}) = 0$, where $\mathcal{F}_n = \sigma(X_m, m \leq n)$. Let $S_n = X_1 + \cdots + X_n$. Then

$$S_{(n)}/n^{1/2} \Rightarrow \sigma B(\cdot)$$

Remark 8.3.1 This result was discovered independently by Billingsley (1961) and Ibragimov (1963).

Proof $u_n \equiv E(X_n^2 | \mathcal{F}_{n-1})$ can be written as $\varphi(X_{n-1}, X_{n-2}, \dots)$, so Theorem 7.1.3 implies u_n is stationary and ergodic, and the ergodic theorem implies

$$n^{-1} \sum_{m=1}^n u_m \rightarrow Eu_0 = EX_0^2 \quad \text{a.s.}$$

The last conclusion shows that (i) of Theorem 8.2.8 holds. To verify (ii), we observe

$$n^{-1} \sum_{m=1}^n E(X_m^2 1_{(|X_m| > \epsilon \sqrt{n})}) = E(X_0^2 1_{(|X_0| > \epsilon \sqrt{n})}) \rightarrow 0$$

by the dominated convergence theorem. $\square$

We prove our first central limit theorem for stationary sequences by reducing to the martingale case. Our proof is based on Scott (1973), but the key is an idea of Gordin (1969). In some applications of this result (see e.g., Exercise 8.3.1), it is convenient to use σ -fields richer than $\mathcal{F}_m = \sigma(X_n, n \leq m)$. So we adopt abstract conditions that satisfy the demands of martingale theory and ergodic theory:

- (i) $n \rightarrow \mathcal{F}_n$ is increasing, and $X_n \in \mathcal{F}_n$
- (ii) $\theta^{-1}\mathcal{F}_m = \mathcal{F}_{m+1}$

Theorem 8.3.2 Suppose $X_n, n \in \mathbf{Z}$ is an ergodic stationary sequence with $EX_n = 0$. Let $\mathcal{F}_n$ satisfy (i), (ii), and

$$\sum_{n \geq 1} \|E(X_0 | \mathcal{F}_{-n})\|_2 < \infty$$

where $\|Y\|_2 = (EY^2)^{1/2}$. Let $S_n = X_1 + \cdots + X_n$. Then $S_{(n\cdot)}/\sqrt{n} \Rightarrow \sigma B(\cdot)$, where

$$\sigma^2 = EX_0^2 + 2 \sum_{n=1}^{\infty} EX_0 X_n$$

and the series in the definition converges absolutely.

Proof Suppose $X_n, n \in \mathbf{Z}$ is defined on sequence space $(\mathbf{R}^{\mathbf{Z}}, \mathcal{R}^{\mathbf{Z}}, P)$ with $X_n(\omega) = \omega_n$ and let $(\theta^n \omega)(m) = \omega(m+n)$. Let

$$\begin{aligned} H_n &= \{Y \in \mathcal{F}_n \text{ with } EY^2 < \infty\} \\ K_n &= \{Y \in H_n \text{ with } E(YZ) = 0 \text{ for all } Z \in H_{n-1}\} \end{aligned}$$

Geometrically, $H_0 \supset H_{-1} \supset H_{-2} \dots$ is a sequence of subspaces of L^2 and K_n is the orthogonal complement of H_{n-1} in H_n . If Y is a random variable, let $(\theta^n Y)(\omega) = Y(\theta^n \omega)$. Generalizing from the example $Y = f(X_{-j}, \dots, X_k)$, which has $\theta^n Y = f(X_{n-j}, \dots, X_{n+k})$, it is easy to see that if $Y \in H_k$, then $\theta^n Y \in H_{k+n}$, and hence if $Y \in K_j$, then $\theta^n Y \in K_{n+j}$.

If X_0 happened to be in K_0 , then we would be happy since then $X_n = \theta^n X_0 \in K_n$ for all n , and taking $Z = 1_A \in H_{n-1}$ we would have $E(X_n 1_A) = 0$ for all $A \in \mathcal{F}_{n-1}$ and hence $E(X_n | \mathcal{F}_{n-1}) = 0$. The next best thing to having $X_0 \in K_0$ is to have

$$X_0 = Y_0 + Z_0 - \theta Z_0 \tag{8.3.1}$$

with $Y_0 \in K_0$ and $Z_0 \in L^2$, for then if we let

$$S_n = \sum_{m=1}^n X_m = \sum_{m=1}^n \theta^m X_0 \quad \text{and} \quad T_n = \sum_{m=1}^n \theta^m Y_0$$

then $S_n = T_n + \theta Z_0 - \theta^{n+1} Z_0$. The $\theta^m Y_0$ are a stationary ergodic martingale difference sequence (ergodicity follows from Theorem 7.1.3), so Theorem 8.3.1 implies

$$T_{(n\cdot)}/\sqrt{n} \Rightarrow \sigma B(\cdot) \quad \text{where } \sigma^2 = EY_0^2$$

To get rid of the other term, we observe $\theta Z_0/\sqrt{n} \rightarrow 0$ a.s. and

$$P\left(\sup_{1 \leq m \leq n} \theta^{m+1} Z_0 > \epsilon \sqrt{n}\right) \leq n P(Z_0 > \epsilon \sqrt{n}) \leq \epsilon^{-2} E(Z_0^2; Z_0 > \epsilon \sqrt{n}) \rightarrow 0$$

by dominated convergence. To solve (8.3.1) formally, we let

$$\begin{aligned} Z_0 &= \sum_{j=0}^{\infty} E(X_j | \mathcal{F}_{-1}) \\ \theta Z_0 &= \sum_{j=0}^{\infty} E(X_{j+1} | \mathcal{F}_0) \\ Y_0 &= \sum_{j=0}^{\infty} \{E(X_j | \mathcal{F}_0) - E(X_j | \mathcal{F}_{-1})\} \end{aligned}$$

and check that

$$Y_0 + Z_0 - \theta Z_0 = E(X_0 | \mathcal{F}_0) = X_0$$

To justify the last calculation, we need to know that the series in the definitions of Z_0 and Y_0 converge. Our assumption and shift invariance imply

$$\sum_{j=0}^{\infty} \|E(X_j | \mathcal{F}_{-1})\|_2 < \infty$$

so the triangle inequality implies that the series for Z_0 converges in L^2 . Since

$$\|E(X_j | \mathcal{F}_0) - E(X_j | \mathcal{F}_{-1})\|_2 \leq \|E(X_j | \mathcal{F}_0)\|_2$$

(the left-hand side is the projection onto $K_0 \subset H_0$), the series for Y_0 also converges in L^2 . Putting the pieces together, we have shown Theorem 8.3.2 with $\sigma^2 = EY_0^2$. To get the indicated formula for σ^2 , observe that conditioning and using Cauchy-Schwarz

$$|EX_0 X_n| = |E(X_0 E(X_n | \mathcal{F}_0))| \leq \|X_0\|_2 \|E(X_n | \mathcal{F}_0)\|_2$$

Shift invariance implies $\|E(X_n | \mathcal{F}_0)\|_2 = \|E(X_0 | \mathcal{F}_{-n})\|_2$, so the series converges absolutely.

$$ES_n^2 = \sum_{j=1}^n \sum_{k=1}^n EX_j X_k = n EX_0^2 + 2 \sum_{m=1}^{n-1} (n-m) EX_0 X_m$$

From this, it follows easily that

$$n^{-1} ES_n^2 \rightarrow EX_0^2 + 2 \sum_{m=1}^{\infty} EX_0 X_m$$

To finish the proof, let $T_n = \sum_{m=1}^n \theta^m Y_0$, observe $\sigma^2 = EY_0^2$ and

$$n^{-1} E(S_n - T_n)^2 = n^{-1} E(\theta Z_0 - \theta^{n+1} Z_0)^2 \leq 4EZ_0^2/n \rightarrow 0$$

since $(a - b)^2 \leq (2a)^2 + (2b)^2$. □

We turn now to examples. In the first one, it is trivial to check the hypothesis of Theorem 8.3.2.

Example 8.3.3 (M-dependent sequences) Let $X_n, n \in \mathbf{Z}$, be a stationary sequence with $EX_n = 0, EX_n^2 < \infty$, and suppose $\{X_j, j \leq 0\}$ and $\{X_k, k > M\}$ are independent. In this case, $E(X_0|\mathcal{F}_{-n}) = 0$ for $n > M$, so Theorem 8.3.2 implies $S_{n,(n\cdot)}/\sqrt{n} \Rightarrow \sigma B(\cdot)$, where

$$\sigma^2 = EX_0^2 + 2 \sum_{m=1}^M EX_0 X_m$$

Remark 8.3.2 For a bare-hands approach to this problem, see theorem 7.3.1 in Chung (1974).

Exercise 8.3.1 Consider the special case of Example 8.3.3 in which $\xi_i, i \in \mathbf{Z}$, are i.i.d. and take values H and T with equal probability. Let $\eta_n = f(\xi_n, \xi_{n+1})$, where $f(H, T) = 1$ and $f(i, j) = 0$ otherwise. $S_n = \eta_1 + \cdots + \eta_n$ counts the number of head runs in $\xi_1, \dots, \xi_{n+1}$. Apply Theorem 8.3.2 with $\mathcal{F}_n = \sigma(X_m : m \leq n+1)$ to show that there are constants μ and σ so that $(S_n - n\mu)/\sigma n^{1/2} \Rightarrow \chi$. What is the random variable Y_0 constructed in the proof of Theorem 8.3.2 in this case?

Example 8.3.4 (Markov chains) Let $\zeta_n, n \in \mathbf{Z}$ be an irreducible Markov chain on a countable state space S in which each ζ_n has the stationary distribution π . Let $X_n = f(\zeta_n)$, where $\sum f(x)\pi(x) = 0$ and $\sum f(x)^2\pi(x) < \infty$. Results in Chapter 6 imply that X_n is an ergodic stationary sequence. If we let $\mathcal{F}_{-n} = \sigma(\zeta_m, m \leq -n)$, then

$$E(X_0|\mathcal{F}_{-n}) = \sum_y p^n(\zeta_{-n}, y) f(y)$$

where $p^n(x, y)$ is the n step transition probability, so

$$\|E(X_0|\mathcal{F}_{-n})\|_2^2 = \sum_x \pi(x) \left(\sum_y p^n(x, y) f(y) \right)^2$$

When f is bounded, we can use $\sum f(x)\pi(x) = 0$ to get the following bound,

$$\|E(X_0|\mathcal{F}_{-n})\|_2^2 \leq \|f\|_\infty \sum_x \pi(x) \|p^n(x, \cdot) - \pi(\cdot)\|^2$$

where $\|f\|_\infty = \sup |f(x)|$ and $\|\cdot\|$ is the total variation norm.

When S is finite, all f are bounded. If the chain is aperiodic, Exercise 5.6 in Chapter 5 implies that

$$\sup_x \|p^n(x, \cdot) - \pi(\cdot)\| \leq C e^{-\epsilon n}$$

and the hypothesis of Theorem 8.3.2 is satisfied. To see that the limiting variance σ^2 may be 0 in this case, consider the modification of Exercise 7.1 in which $X_n = (\xi_n, \xi_{n+1})$, $f(H, T) = 1, f(T, H) = -1$ and $f(H, H) = f(T, T) = 0$. In this case, $\sum_{m=1}^n f(X_m) \in \{-1, 0, 1\}$ so there is no central limit theorem.

Our last example is simple to treat directly, but will help us evaluate the strength of the conditions in Theorem 8.3.2 and in later theorems.

Example 8.3.5 (Moving average process) Suppose

$$X_m = \sum_{k \geq 0} c_k \xi_{m-k} \quad \text{where} \quad \sum_{k \geq 0} c_k^2 < \infty$$

and the $\xi_i, i \in \mathbf{Z}$, are i.i.d. with $E\xi_i = 0$ and $E\xi_i^2 = 1$. If $\mathcal{F}_{-n} = \sigma(\xi_m; m \leq -n)$, then

$$\|E(X_0|\mathcal{F}_{-n})\|_2 = \left\| \sum_{k \geq n} c_k \xi_{-k} \right\|_2 = \left(\sum_{k \geq n} c_k^2 \right)^{1/2}$$

If, for example, $c_k = (1+k)^{-p}$, $\|E(X_0|\mathcal{F}_{-n})\|_2 \sim n^{(1/2)-p}$, and Theorem 8.3.5 applies if $p > 3/2$.

Remark 8.3.2 Theorem 5.3 in Hall and Heyde (1980) shows that

$$\sum_{j \geq 0} \|E(X_j|\mathcal{F}_0) - E(X_j|\mathcal{F}_{-1})\|_2 < \infty$$

is sufficient for a central limit theorem. Using this result shows that $\sum |c_k| < \infty$ is sufficient for the central limit theorem in Example 7.3.

The condition in the improved result is close to the best possible. Suppose the ξ_i take values 1 and -1 with equal probability and let $c_k = (1+k)^{-p}$, where $1/2 < p < 1$. The Lindeberg-Feller theorem can be used to show that $S_n/n^{3/2-p} \Rightarrow \sigma\chi$. To check the normalization, note that

$$\sum_{m=1}^n X_m = \sum_{j \leq n} a_{n,j} \xi_j$$

If $j \geq 0$, then $a_{n,j} = \sum_{i=0}^{n-j} c_k \approx (n-j)^{1-p}$, so using $0 \leq j \leq n/2$ the variance is at least n^{3-2p} . Further details are left to the reader.

8.3.1 Mixing Properties

The last three examples show that in many cases it is easy to verify the hypothesis of Theorem 8.3.2 directly. To connect Theorem 8.3.2 with other results in the literature, we will introduce sufficient conditions phrased in terms of quantitative measures of dependence between σ -fields. Our first is

$$\alpha(\mathcal{G}, \mathcal{H}) = \sup\{|P(A \cap B) - P(A)P(B)| : A \in \mathcal{G}, B \in \mathcal{H}\}$$

If $\alpha = 0$, $\mathcal{G}$ and $\mathcal{H}$ are independent, so α measures the dependence between the σ -fields.

Lemma 8.3.6 *Let $p, q, r \in (1, \infty]$ with $1/p + 1/q + 1/r = 1$, and suppose $X \in \mathcal{G}, Y \in \mathcal{H}$ have $E|X|^p, E|Y|^q < \infty$. Then*

$$|EXY - EXEY| \leq 8\|X\|_p\|Y\|_q(\alpha(\mathcal{G}, \mathcal{H}))^{1/r}$$

Here, we interpret $x^0 = 1$ for $x > 0$ and $0^0 = 0$.

Proof If $\alpha = 0$, X and Y are independent and the result is true, so we can suppose $\alpha > 0$. We build up to the result in three steps, starting with the case $r = \infty$.

$$|EXY - EXEY| \leq 2\|X\|_p\|Y\|_q \quad (8.3.2)$$

Proof Hölder's inequality implies $|EXY| \leq \|X\|_p\|Y\|_q$, and Jensen's inequality implies

$$\|X\|_p\|Y\|_q \geq |E|X||E|Y|| \geq |EXEY|$$

so the result follows from the triangle inequality. $\square$

$$|EXY - EXEY| \leq 4\|X\|_\infty\|Y\|_\infty\alpha(\mathcal{G}, \mathcal{H}) \quad (8.3.3)$$

Proof Let $\eta = \text{sgn}\{E(Y|\mathcal{G}) - EY\} \in \mathcal{G}$. $EXY = E(XE(Y|\mathcal{G}))$, so

$$\begin{aligned} |EXY - EXEY| &= |E(X\{E(Y|\mathcal{G}) - EY\})| \\ &\leq \|X\|_\infty E|E(Y|\mathcal{G}) - EY| \\ &= \|X\|_\infty E(\eta\{E(Y|\mathcal{G}) - EY\}) \\ &= \|X\|_\infty \{E(\eta Y) - E\eta EY\} \end{aligned}$$

Applying the last result with $X = Y$ and $Y = \eta$ gives

$$|E(Y\eta) - EYE\eta| \leq \|Y\|_\infty |E(\zeta\eta) - E\zeta E\eta|$$

where $\zeta = \text{sgn}\{E(\eta|\mathcal{H}) - E\eta\}$. Now $\eta = 1_A - 1_B$ and $\zeta = 1_C - 1_D$, so

$$\begin{aligned} |E(\zeta\eta) - E\zeta E\eta| &= |P(A \cap C) - P(B \cap C) - P(A \cap D) + P(B \cap D) \\ &\quad - P(A)P(C) + P(B)P(C) + P(A)P(D) - P(B)P(D)| \\ &\leq 4\alpha(\mathcal{G}, \mathcal{H}) \end{aligned}$$

Combining the last three displays gives the desired result. $\square$

$$|EXY - EXEY| \leq 6\|X\|_p\|Y\|_\infty\alpha(\mathcal{G}, \mathcal{H})^{1-1/p} \quad (8.3.4)$$

Proof Let $C = \alpha^{-1/p}\|X\|_p$, $X_1 = X1_{(|X| \leq C)}$ and $X_2 = X - X_1$

$$\begin{aligned} |EXY - EXEY| &\leq |EX_1Y - EX_1EY| + |EX_2Y - EX_2EY| \\ &\leq 4\alpha C\|Y\|_\infty + 2\|Y\|_\infty E|X_2| \end{aligned}$$

by (8.3.3) and (8.3.2). Now

$$E|X_2| \leq C^{-(p-1)}E(|X|^p 1_{(|X| > C)}) \leq C^{-p+1}E|X|^p$$

Combining the last two inequalities and using the definition of C gives

$$|EXY - EXEY| \leq 4\alpha^{1-1/p}\|X\|_p\|Y\|_\infty + 2\|Y\|_\infty\alpha^{1-1/p}\|X\|_p^{-p+1+p}$$

which proves (8.3.4) $\square$

Finally, to prove Lemma 8.3.6, let $C = \alpha^{-1/q} \|Y\|_q$, $Y_1 = Y 1_{(|Y| \leq C)}$, and $Y_2 = Y - Y_1$.

$$\begin{aligned} |EXY - EXEY| &\leq |EXY_1 - EXEY_1| + |EXY_2 - EXEY_2| \\ &\leq 6C \|X\|_p \alpha^{1-1/p} + 2\|X\|_p \|Y_2\|_\theta \end{aligned}$$

where $\theta = (1 - 1/p)^{-1}$ by (8.3.4) and (8.3.2). Now

$$E|Y_2|^\theta \leq C^{-q+\theta} E(|Y|^q 1_{(|Y|>C)}) \leq C^{-q+\theta} E|Y|^q$$

Taking the $1/\theta$ root of each side and recalling the definition of C

$$\|Y_2\|_\theta \leq C^{-(q-\theta)/\theta} \|Y\|_q^{q/\theta} \leq \alpha^{(q-\theta)/\theta q} \|Y\|_q$$

so we have

$$|EXY - EXEY| \leq 6\alpha^{-1/q} \|Y\|_q \|X\|_p \alpha^{1-1/p} + 2\|X\|_p \alpha^{1/\theta-1/q} \|Y\|_q^{1/\theta+1/q}$$

proving the desired result. $\square$

Remark 8.3.3 The last proof is from appendix III of Hall and Heyde (1980). They attribute (b) to Ibragimov (1962) and (c) and Lemma 8.3.6 to Davydov (1968).

Theorem 8.3.7 Suppose X_n , $n \in \mathbf{Z}$ is an ergodic stationary sequence with $EX_n = 0$, $E|X_0|^{2+\delta} < \infty$. Let $\alpha(n) = \alpha(\mathcal{F}_{-n}, \sigma(X_0))$, where $\mathcal{F}_{-n} = \sigma(X_m : m \leq -n)$, and suppose

$$\sum_{n=1}^{\infty} \alpha(n)^{\delta/2(2+\delta)} < \infty$$

If $S_n = X_1 + \cdots + X_n$, then $S_{(n)}/\sqrt{n} \Rightarrow \sigma B(\cdot)$, where

$$\sigma^2 = EX_0^2 + 2 \sum_{n=1}^{\infty} EX_0 X_n$$

Remark If one strengthens the mixing condition, one can weaken the summability condition. Let $\bar{\alpha}(n) = \alpha(\mathcal{F}_{-n}, \mathcal{F}'_0)$, where $\mathcal{F}'_0 = \sigma(X_k, k \geq 0)$. When $\bar{\alpha}(n) \downarrow 0$, the sequence is called **strong mixing**. Rosenblatt (1956) introduced the concept as a condition under which the central limit theorem for stationary sequences could be obtained. Ibragimov (1962) proved $S_n/\sqrt{n} \Rightarrow \sigma \chi$, where $\sigma^2 = \lim_{n \rightarrow \infty} ES_n^2/n$ under the assumption

$$\sum_{n=1}^{\infty} \bar{\alpha}(n)^{\delta/(2+\delta)} < \infty$$

See Ibragimov and Linnik (1971), Theorem 18.5.3, or Hall and Heyde (1980), Corollary 5.1, for a proof.

Proof To use Lemma 8.3.6 to estimate the quantity in Theorem 8.3.2, we begin with

$$\|E(X|\mathcal{F})\|_2 = \sup\{E(XY) : Y \in \mathcal{F}, \|Y\|_2 = 1\} \quad (8.3.5)$$

Proof of (8.3.5) If $Y \in \mathcal{F}$ with $\|Y\|_2 = 1$, then using a by now familiar property of conditional expectation and the Cauchy-Schwarz inequality

$$EXY = E(E(XY|\mathcal{F})) = E(YE(X|\mathcal{F})) \leq \|E(X|\mathcal{F})\|_2 \|Y\|_2$$

Equality holds when $Y = E(X|\mathcal{F})/\|E(X|\mathcal{F})\|_2$. □

Letting $p = 2 + \delta$ and $q = 2$ in Lemma 8.3.6, noticing

$$\frac{1}{r} = 1 - \frac{1}{p} - \frac{1}{q} = 1 - \frac{1}{2+\delta} - \frac{1}{2} = \frac{4+2\delta-2-(2+\delta)}{(2+\delta)2} = \frac{\delta}{(2+\delta)2}$$

and recalling $EX_0 = 0$, shows that if $Y \in \mathcal{F}_{-n}$

$$|EX_0 Y| \leq 8\|X_0\|_{2+\delta}\|Y\|_2 \alpha(n)^{\delta/2(2+\delta)}$$

Combining this with (8.3.5) gives

$$\|E(X_0|\mathcal{F}_{-n})\|_2 \leq 8\|X_0\|_{2+\delta} \alpha(n)^{\delta/2(2+\delta)}$$

and it follows that the hypotheses of Theorem 8.3.2 are satisfied. □

In the M dependent case (Example 8.3.3), $\alpha(n) = 0$ for $n > M$, so Theorem 8.3.7 applies. As for Markov chains (Example 8.3.4), in this case

$$\begin{aligned} \alpha(n) &= \sup_{A,B} |P(X_{-n} \in A, X_0 \in B) - \pi(A)\pi(B)| \\ &= \sup_{A,B} \left| \sum_{x \in A, y \in B} \pi(x) p^n(x, y) - \pi(x)\pi(y) \right| \\ &\leq \sum_x \pi(x) 2\|p^n(x, \cdot) - \pi(\cdot)\| \end{aligned}$$

so the hypothesis of Theorem 8.3.7 can be checked if we know enough about the rate of convergence to equilibrium.

Finally, to see how good the conditions in Theorem 8.3.7 are, we consider the special case of the moving average process (Example 8.3.5) in which the ξ_i are i.i.d. standard normals. Let

$$\rho(\mathcal{G}, \mathcal{H}) = \sup\{\text{corr}(X, Y) : X \in \mathcal{G}, Y \in \mathcal{H}\}$$

where

$$\text{corr}(X, Y) = \frac{EXY - EXEY}{\|X - EX\|_2 \|Y - EY\|_2}$$

Clearly, $\alpha(\mathcal{G}, \mathcal{H}) \leq \rho(\mathcal{G}, \mathcal{H})$. Kolmogorov and Rozanov (1960) have shown that when $\mathcal{G}$ and $\mathcal{H}$ are generated by Gaussian random variables, then $\rho(\mathcal{G}, \mathcal{H}) \leq 2\pi\alpha(\mathcal{G}, \mathcal{H})$. They proved this by showing that $\rho(\mathcal{G}, \mathcal{H})$ is the angle between $L^2(\mathcal{G})$ and $L^2(\mathcal{H})$. Using the geometric interpretation, we see that if $|c_k|$ is decreasing, then $\alpha(n) \leq \bar{\alpha}(n) \leq |c_n|$, so Theorem 8.3.7 requires $\sum |c_n|^{(1/2)-\epsilon} < \infty$, but Ibragimov's result applies if $\sum |c_n|^{1-\epsilon} < \infty$. However, as remarked earlier, the central limit theorem is valid if $\sum |c_n| < \infty$.

8.4 Empirical Distributions, Brownian Bridge

Let $X_1, X_2, \dots$ be i.i.d. with distribution F . Let

$$\hat{F}_n(x) = \frac{1}{n} |\{m \leq n : X_m \leq x\}|$$

be the empirical distribution. In Theorem 2.4.9 we showed

Theorem 8.4.1 (The Glivenko-Cantelli theorem) As $n \rightarrow \infty$,

$$\sup_x |F_n(x) - F(x)| \rightarrow 0 \quad a.s.$$

In this section, we will investigate the rate of convergence when F is continuous. We impose this restriction, so we can reduce to the case of a uniform distribution on $(0,1)$ by setting $Y_n = F(X_n)$. Since $x \rightarrow F(x)$ is nondecreasing and continuous and no observations land in intervals of constancy of F , it is easy to see that if we let

$$\hat{G}_n(y) = \frac{1}{n} |\{m \leq n : Y_m \leq y\}|$$

then

$$\sup_x |\hat{F}_n(x) - F(x)| = \sup_{0 < y < 1} |\hat{G}_n(y) - y|$$

For the rest of the section then, we will assume $Y_1, Y_2, \dots$ is i.i.d. uniform on $(0,1)$. To be able to apply Donsker's theorem, we will transform the problem. Put the observations $Y_1, \dots, Y_n$ in increasing order: $U_1^n < U_2^n < \dots < U_n^n$. I claim that

$$\begin{aligned} \sup_{0 < y < 1} \hat{G}_n(y) - y &= \sup_{1 \leq m \leq n} \frac{m}{n} - U_m^n \\ \inf_{0 < y < 1} \hat{G}_n(y) - y &= \inf_{1 \leq m \leq n} \frac{m-1}{n} - U_m^n \end{aligned} \quad (8.4.1)$$

since the sup occurs at a jump of $\hat{G}_n$ and the inf right before a jump. For a picture, see Figure 8.1. We will show that

$$D_n \equiv n^{1/2} \sup_{0 < y < 1} |\hat{G}_n(y) - y|$$

has a limit, so the extra $-1/n$ in the inf does not make any difference.

Our third and final maneuver is to give a special construction of the order statistics $U_1^n < U_2^n < \dots < U_n^n$. Let $W_1, W_2, \dots$ be i.i.d. with $P(W_i > t) = e^{-t}$ and let $Z_n = W_1 + \dots + W_n$.

Lemma 8.4.2 $\{U_k^n : 1 \leq k \leq n\} \stackrel{d}{=} \{Z_k/Z_{n+1} : 1 \leq k \leq n\}$

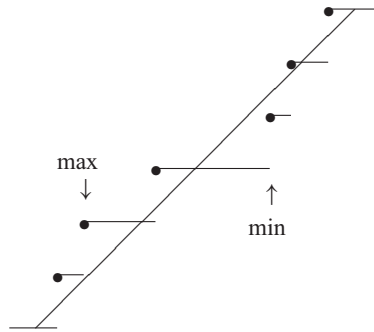


Figure 8.1 Picture proof of formulas in (8.4.1).

Proof We change variables $v = r(t)$, where $v_i = t_i/t_{n+1}$ for $i \leq n$, $v_{n+1} = t_{n+1}$. The inverse function is

$$s(v) = (v_1 v_{n+1}, \dots, v_n v_{n+1}, v_{n+1})$$

which has matrix of partial derivatives $\partial s_i / \partial v_j$ given by

$$\begin{pmatrix} v_{n+1} & 0 & \dots & 0 & v_1 \\ 0 & v_{n+1} & \dots & 0 & v_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & v_{n+1} & v_n \\ 0 & 0 & \dots & 0 & 1 \end{pmatrix}$$

The determinant of this matrix is v_{n+1}^n , so if we let $W = (V_1, \dots, V_{n+1}) = r(Z_1, \dots, Z_{n+1})$, the change of variables formula implies W has joint density

$$f_W(v_1, \dots, v_n, v_{n+1}) = \left(\prod_{m=1}^n \lambda e^{-\lambda v_{n+1}(v_m - v_{m-1})} \right) \lambda e^{-\lambda v_{n+1}(1 - v_n)} v_{n+1}^n$$

To find the joint density of $V = (V_1, \dots, V_n)$, we simplify the preceding formula and integrate out the last coordinate to get

$$f_V(v_1, \dots, v_n) = \int_0^\infty \lambda^{n+1} v_{n+1}^n e^{-\lambda v_{n+1}} dv_{n+1} = n!$$

for $0 < v_1 < v_2 < \dots < v_n < 1$, which is the desired joint density. $\square$

We turn now to the limit law for D_n . As argued earlier, it suffices to consider

$$\begin{aligned} D'_n &= n^{1/2} \max_{1 \leq m \leq n} \left| \frac{Z_m}{Z_{n+1}} - \frac{m}{n} \right| \\ &= \frac{n}{Z_{n+1}} \max_{1 \leq m \leq n} \left| \frac{Z_m}{n^{1/2}} - \frac{m}{n} \cdot \frac{Z_{n+1}}{n^{1/2}} \right| \\ &= \frac{n}{Z_{n+1}} \max_{1 \leq m \leq n} \left| \frac{Z_m - m}{n^{1/2}} - \frac{m}{n} \cdot \frac{Z_{n+1} - n}{n^{1/2}} \right| \end{aligned} \quad (8.4.2)$$

If we let

$$B_n(t) = \begin{cases} (Z_m - m)/n^{1/2} & \text{if } t = m/n \text{ with } m \in \{0, 1, \dots, n\} \\ \text{linear} & \text{on } [(m-1)/n, m/n] \end{cases}$$

then

$$D'_n = \frac{n}{Z_{n+1}} \max_{0 \leq t \leq 1} \left| B_n(t) - t \left\{ B_n(1) + \frac{Z_{n+1} - Z_n}{n^{1/2}} \right\} \right|$$

The strong law of large numbers implies $Z_{n+1}/n \rightarrow 1$ a.s., so the first factor will disappear in the limit. To find the limit of the second, we observe that Donsker's theorem, Theorem 8.1.4, implies $B_n(\cdot) \Rightarrow B(\cdot)$, a Brownian motion, and computing second moments shows

$$(Z_{n+1} - Z_n)/n^{1/2} \rightarrow 0 \text{ in probability}$$

$\psi(\omega) = \max_{0 \leq t \leq 1} |\omega(t) - t\omega(1)|$ is a continuous function from $C[0, 1]$ to $\mathbf{R}$, so it follows from Donsker's theorem that:

Theorem 8.4.3 $D_n \Rightarrow \max_{0 \leq t \leq 1} |B_t - tB_1|$, where B_t is a Brownian motion starting at 0.

Remark Doob (1949) suggested this approach to deriving results of Kolmogorov and Smirnov, which was later justified by Donsker (1952). Our proof follows Breiman (1968).

To identify the distribution of the limit in Theorem 8.4.3, we will first prove

$$\{B_t - tB_1, 0 \leq t \leq 1\} \stackrel{d}{=} \{B_t, 0 \leq t \leq 1 | B_1 = 0\} \quad (8.4.3)$$

a process we will denote by B_t^0 and call the **Brownian bridge**. The event $B_1 = 0$ has probability 0, but it is easy to see what the conditional probability should mean. If $0 = t_0 < t_1 < \dots < t_n < t_{n+1} = 1$, $x_0 = 0$, $x_{n+1} = 0$, and $x_1, \dots, x_n \in \mathbf{R}$, then

$$\begin{aligned} P(B(t_1) = x_1, \dots, B(t_n) = x_n | B(1) = 0) \\ = \frac{1}{p_1(0, 0)} \prod_{m=1}^{n+1} p_{t_m - t_{m-1}}(x_{m-1}, x_m) \end{aligned} \quad (8.4.4)$$

where $p_t(x, y) = (2\pi t)^{-1/2} \exp(-(y - x)^2/2t)$.

Proof of (8.4.3) Formula (8.4.4) shows that the f.d.d.'s of B_t^0 are multivariate normal and have mean 0. Since $B_t - tB_1$ also has this property, it suffices to show that the covariances are equal. We begin with the easier computation. If $s < t$, then

$$E((B_s - sB_1)(B_t - tB_1)) = s - st - st + st = s(1 - t) \quad (8.4.5)$$

For the other process, $P(B_s^0 = x, B_t^0 = y)$ is

$$\begin{aligned} \frac{\exp(-x^2/2s)}{(2\pi s)^{1/2}} \cdot \frac{\exp(-(y - x)^2/2(t - s))}{(2\pi(t - s))^{1/2}} \cdot \frac{\exp(-y^2/2(1 - t))}{(2\pi(1 - t))^{1/2}} \cdot (2\pi)^{1/2} \\ = (2\pi)^{-1} (s(t - s)(1 - t))^{-1/2} \exp(-(ax^2 + 2bxy + cy^2)/2) \end{aligned} \quad (8.4.6)$$

where

$$\begin{aligned} a &= \frac{1}{s} + \frac{1}{t - s} = \frac{t}{s(t - s)} & b &= -\frac{1}{t - s} \\ c &= \frac{1}{t - s} + \frac{1}{1 - t} = \frac{1 - s}{(t - s)(1 - t)} \end{aligned}$$

Recalling the discussion at the end of Section 3.9 and noticing

$$\begin{pmatrix} \frac{t}{s(t-s)} & \frac{-1}{(t-s)} \\ \frac{-1}{(t-s)} & \frac{1-s}{(t-s)(1-t)} \end{pmatrix}^{-1} = \begin{pmatrix} s(1-s) & s(1-t) \\ s(1-t) & t(1-t) \end{pmatrix}$$

(multiply the matrices!) shows (8.4.3) holds. $\square$

Remark To do computations with the joint density, it is useful to know that if (X, Y) have a bivariate normal distribution with mean 0 $EX^2 = \sigma_X^2$, $EY^2 = \sigma_Y^2$, and correlation ρ , then the joint density is given by

$$\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left(\frac{x^2}{\sigma_X^2} - \frac{2xy}{\sigma_X\sigma_Y} + \frac{y^2}{\sigma_Y^2}\right)\right]$$

In (8.4.6) if $X = B_s^0$ and $Y = B_t^0$, then $\sigma_X = \sqrt{s(1-s)}$, $\sigma_Y = \sqrt{t(1-t)}$ and

$$\rho = \frac{s(1-t)}{\sqrt{s(1-s)t(1-t)}} = \sqrt{\frac{s(1-t)}{(1-s)t}}$$

so we have

$$1 - \rho^2 = \frac{(1-s)t - s(1-t)}{(1-s)t} = \frac{t-s}{(1-s)t}.$$

We leave it to the reader to check the well-known fact that the conditional distribution

$$(Y|X = x) = \text{normal}(x\rho\sigma_Y/\sigma_X, (1-\rho^2)\sigma_Y^2)$$

which in our special case implies

$$(B_t^0|B_s^0 = x) = \text{normal}(x(1-t)/(1-s), (1-t)(1-s)/(1-s)) \quad (8.4.7)$$

Our final step in investigating the limit distribution of D_n is to compute the distribution of $\max_{0 \leq t \leq 1} |B_t^0|$. To do this, we first prove

Theorem 8.4.4 *The density function of B_t on $\{T_a \wedge T_b > t\}$ is*

$$\begin{aligned} P_x(T_a \wedge T_b > t, B_t = y) &= \sum_{n=-\infty}^{\infty} P_x(B_t = y + 2n(b-a)) \\ &\quad - P_x(B_t = 2b - y + 2n(b-a)) \end{aligned} \quad (8.4.8)$$

Proof We begin by observing that if $A \subset (a, b)$

$$\begin{aligned} P_x(T_a \wedge T_b > t, B_t \in A) &= P_x(B_t \in A) - P_x(T_a < T_b, T_a < t, B_t \in A) \\ &\quad - P_x(T_b < T_a, T_b < t, B_t \in A) \end{aligned} \quad (8.4.9)$$

If we let $\rho_a(y) = 2a - y$ be reflection through a and observe that $\{T_a < T_b\}$ is $\mathcal{F}(T_a)$ measurable, then it follows from the proof of (7.4.5) that

$$P_x(T_a < T_b, T_a < t, B_t \in A) = P_x(T_a < T_b, B_t \in \rho_a A)$$

where $\rho_a A = \{\rho_a(y) : y \in A\}$. To get rid of the $T_a < T_b$, we observe that

$$P_x(T_a < T_b, B_t \in \rho_a A) = P_x(B_t \in \rho_a A) - P_x(T_b < T_a, B_t \in \rho_a A)$$

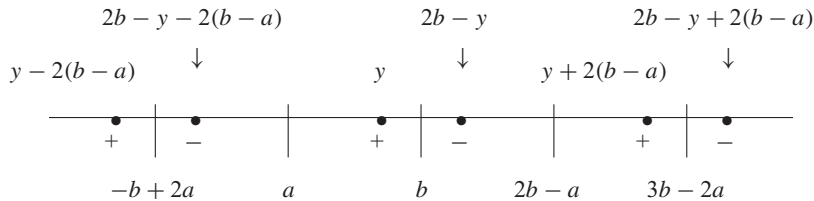


Figure 8.2 Picture of the infinite series in (8.4.8). Note that the array of + and - is anti-symmetric when seen from a or b .

Noticing that $B_t \in \rho_a A$ and $T_b < T_a$ imply $T_b < t$ and using the reflection principle again gives

$$\begin{aligned} P_x(T_b < T_a, B_t \in \rho_a A) &= P_x(T_b < T_a, B_t \in \rho_b \rho_a A) \\ &= P_x(B_t \in \rho_b \rho_a A) - P_x(T_a < T_b, B_t \in \rho_b \rho_a A) \end{aligned}$$

Repeating the last two calculations n more times gives

$$\begin{aligned} P_x(T_a < T_b, B_t \in \rho_a A) &= \sum_{m=0}^n P_x(B_t \in \rho_a (\rho_b \rho_a)^m A) - P_x(B_t \in (\rho_b \rho_a)^{m+1} A) \\ &\quad + P_x(T_a < T_b, B_t \in (\rho_b \rho_a)^{n+1} A) \end{aligned}$$

Each pair of reflections pushes A further away from 0, so letting $n \rightarrow \infty$ shows

$$P_x(T_a < T_b, B_t \in \rho_a A) = \sum_{m=0}^{\infty} P_x(B_t \in \rho_a (\rho_b \rho_a)^m A) - P_x(B_t \in (\rho_b \rho_a)^{m+1} A)$$

Interchanging the roles of a and b gives

$$P_x(T_b < T_a, B_t \in \rho_b A) = \sum_{m=0}^{\infty} P_x(B_t \in \rho_b (\rho_a \rho_b)^m A) - P_x(B_t \in (\rho_a \rho_b)^{m+1} A)$$

Combining the last two expressions with (8.4.9) and using $\rho_c^{-1} = \rho_c$, $(\rho_a \rho_b)^{-1} = \rho_b^{-1} \rho_a^{-1}$ gives

$$P_x(T_a \wedge T_b > t, B_t \in A) = \sum_{m=-\infty}^{\infty} P_x(B_t \in (\rho_b \rho_a)^n A) - P_x(B_t \in \rho_a (\rho_b \rho_a)^n A)$$

To prepare for applications, let $A = (u, v)$, where $a < u < v < b$, notice that $\rho_b \rho_a(y) = y + 2(b - a)$, and change variables in the second sum to get

$$\begin{aligned} P_x(T_a \wedge T_b > t, u < B_t < v) &= \\ &\sum_{n=-\infty}^{\infty} \{P_x(u + 2n(b - a) < B_t < v + 2n(b - a)) \\ &\quad - P_x(2b - v + 2n(b - a) < B_t < 2b - u + 2n(b - a))\} \end{aligned} \quad (8.4.10)$$

Letting $u = y - \epsilon$, $v = y + \epsilon$, dividing both sides by 2ϵ , and letting $\epsilon \rightarrow 0$ (leaving it to the reader to check that the dominated convergence theorem applies) gives the desired result. $\square$

Setting $x = y = 0$, $t = 1$, and dividing by $(2\pi)^{-1/2} = P_0(B_1 = 0)$, we get a result for the Brownian bridge B_t^0 :

$$\begin{aligned} P_0 \left(a < \min_{0 \leq t \leq 1} B_t^0 < \max_{0 \leq t \leq 1} B_t^0 < b \right) \\ = \sum_{n=-\infty}^{\infty} e^{-(2n(b-a))^2/2} - e^{-(2b+2n(b-a))^2/2} \end{aligned} \quad (8.4.11)$$

Taking $a = -b$, we have

$$P_0 \left(\max_{0 \leq t \leq 1} |B_t^0| < b \right) = \sum_{m=-\infty}^{\infty} (-1)^m e^{-2m^2 b^2} \quad (8.4.12)$$

This formula gives the distribution of the Kolmogorov-Smirnov statistic, which can be used to test if an i.i.d. sequence $X_1, \dots, X_n$ has distribution F . To do this, we transform the data to $F(X_n)$ and look at the maximum discrepancy between the empirical distribution and the uniform. (8.4.12) tells us the distribution of the error when the X_i have distribution F .

Exercises

8.4.1 Use Exercise 7.4.8 and the reasoning that led to (8.4.11) to conclude

$$P \left(\max_{0 \leq t \leq 1} B_t^0 > b \right) = \exp(-2b^2)$$

8.4.2 Let B_t be a Brownian motion starting at 0. Show that $X_t = (1-t)B(t/1-t)$ is a Brownian bridge.

8.4.3 Show that Brownian bridge is a Markov process by checking that if $s_1 < \dots < s_n < s < t < 1$

$$P(B(t) = y | B(s) = x, B(s_n) = x_n, \dots, B(s_1) = x_1) = P(B(t) = y | B(s) = x)$$

8.4.4 Let B_t^0 be Brownian bridge. Show that $X_t = B_t^0/(1-t)$ is a martingale but is not L^2 bounded.

8.5 Laws of the Iterated Logarithm

Our first goal is to show:

Theorem 8.5.1 (LIL for Brownian motion)

$$\limsup_{t \rightarrow \infty} B_t / (2t \log \log t)^{1/2} = 1 \quad a.s.$$

Here LIL is short for “law of the iterated logarithm,” a name that refers to the $\log \log t$ in the denominator. Once Theorem 8.5.1 is established, we can use the Skorokhod representation to prove the analogous result for random walks with mean 0 and finite variance.

Proof The key to the proof is (7.4.4).

$$P_0 \left(\max_{0 \leq s \leq 1} B_s > a \right) = P_0(T_a \leq 1) = 2 P_0(B_1 \geq a) \quad (8.5.1)$$

To bound the right-hand side, we use the result in Lemma 1.2.6. For $x > 0$,

$$(x^{-1} - x^{-3}) \exp(-x^2/2) \leq \int_x^\infty \exp(-y^2/2) dy \leq x^{-1} \exp(-x^2/2) \quad (8.5.2)$$

and hence

$$\int_x^\infty \exp(-y^2/2) dy \sim \frac{1}{x} \exp(-x^2/2) \quad \text{as } x \rightarrow \infty \quad (8.5.3)$$

(8.5.3) and Brownian scaling imply that

$$P_0(B_t > (tf(t))^{1/2}) \sim \kappa f(t)^{-1/2} \exp(-f(t)/2)$$

where $\kappa = (2\pi)^{-1/2}$ is a constant that we will try to ignore. The last result implies that if $\epsilon > 0$, then

$$\sum_{n=1}^{\infty} P_0(B_n > (nf(n))^{1/2}) \begin{cases} < \infty & \text{when } f(n) = (2 + \epsilon) \log n \\ = \infty & \text{when } f(n) = (2 - \epsilon) \log n \end{cases}$$

and hence by the Borel-Cantelli lemma that

$$\limsup_{n \rightarrow \infty} B_n / (2n \log n)^{1/2} \leq 1 \quad \text{a.s.}$$

To replace $\log n$ by $\log \log n$, we have to look along exponentially growing sequences. Let $t_n = \alpha^n$, where $\alpha > 1$.

$$\begin{aligned} P_0 \left(\max_{t_n \leq s \leq t_{n+1}} B_s > (t_n f(t_n))^{1/2} \right) &\leq P_0 \left(\max_{0 \leq s \leq t_{n+1}} B_s / t_{n+1}^{1/2} > \left(\frac{f(t_n)}{\alpha} \right)^{1/2} \right) \\ &\leq 2\kappa (f(t_n)/\alpha)^{-1/2} \exp(-f(t_n)/2\alpha) \end{aligned}$$

by (8.5.1) and (8.5.2). If $f(t) = 2\alpha^2 \log \log t$, then

$$\log \log t_n = \log(n \log \alpha) = \log n + \log \log \alpha$$

so $\exp(-f(t_n)/2\alpha) \leq C_\alpha n^{-\alpha}$, where C_α is a constant that depends only on α , and hence

$$\sum_{n=1}^{\infty} P_0 \left(\max_{t_n \leq s \leq t_{n+1}} B_s > (t_n f(t_n))^{1/2} \right) < \infty$$

Since $t \rightarrow (tf(t))^{1/2}$ is increasing and $\alpha > 1$ is arbitrary, it follows that

$$\limsup B_t / (2t \log \log t)^{1/2} \leq 1 \quad (8.5.4)$$

To prove the other half of Theorem 8.5.1, again let $t_n = \alpha^n$, but this time α will be large, since to get independent events, we will look at

$$P_0 \left(B(t_{n+1}) - B(t_n) > (t_{n+1} f(t_{n+1}))^{1/2} \right) = P_0 \left(B_1 > (\beta f(t_{n+1}))^{1/2} \right)$$

where $\beta = t_{n+1}/(t_{n+1} - t_n) = \alpha/(\alpha - 1) > 1$. The last quantity is

$$\geq \frac{\kappa}{2} (\beta f(t_{n+1}))^{-1/2} \exp(-\beta f(t_{n+1})/2)$$

if n is large by (8.5.3). If $f(t) = (2/\beta^2) \log \log t$, then $\log \log t_n = \log n + \log \log \alpha$, so

$$\exp(-\beta f(t_{n+1})/2) \geq C_\alpha n^{-1/\beta}$$

where C_α is a constant that depends only on α , and hence

$$\sum_{n=1}^{\infty} P_0 \left(B(t_{n+1}) - B(t_n) > (t_{n+1} f(t_{n+1}))^{1/2} \right) = \infty$$

Since the events in question are independent, it follows from the second Borel-Cantelli lemma that

$$B(t_{n+1}) - B(t_n) > ((2/\beta^2)t_{n+1} \log \log t_{n+1})^{1/2} \quad \text{i.o.} \quad (8.5.5)$$

From (8.5.4), we get

$$\limsup_{n \rightarrow \infty} B(t_n)/(2t_n \log \log t_n)^{1/2} \leq 1 \quad (8.5.6)$$

Since $t_n = t_{n+1}/\alpha$ and $t \rightarrow \log \log t$ is increasing, combining (8.5.5) and (8.5.6), and recalling $\beta = \alpha/(\alpha - 1)$ gives

$$\limsup_{n \rightarrow \infty} B(t_{n+1})/(2t_{n+1} \log \log t_{n+1})^{1/2} \geq \frac{\alpha - 1}{\alpha} - \frac{1}{\alpha^{1/2}}$$

Letting $\alpha \rightarrow \infty$ now gives the desired lower bound, and the proof of Theorem 8.5.1 is complete. $\square$

Turning to random walk, we will prove a result due to Hartman and Wintner (1941):

Theorem 8.5.2 *If $X_1, X_2, \dots$ are i.i.d. with $EX_i = 0$ and $EX_i^2 = 1$, then*

$$\limsup_{n \rightarrow \infty} S_n/(2n \log \log n)^{1/2} = 1$$

Proof By Theorem 8.1.2, we can write $S_n = B(T_n)$ with $T_n/n \rightarrow 1$ a.s. As in the proof of Donsker's theorem, this is all we will use in the following argument. Theorem 8.5.2 will follow from Theorem 8.5.1 once we show

$$(S_{[t]} - B_t)/(t \log \log t)^{1/2} \rightarrow 0 \quad \text{a.s.} \quad (8.5.7)$$

To do this, we begin by observing that if $\epsilon > 0$ and $t \geq t_o(\omega)$

$$T_{[t]} \in [t/(1 + \epsilon), t(1 + \epsilon)] \quad (8.5.8)$$

To estimate $S_{[t]} - B_t$, we let $M(t) = \sup\{|B(s) - B(t)| : t/(1 + \epsilon) \leq s \leq t(1 + \epsilon)\}$. To control the last quantity, we let $t_k = (1 + \epsilon)^k$ and notice that if $t_k \leq t \leq t_{k+1}$

$$\begin{aligned} M(t) &\leq \sup\{|B(s) - B(t)| : t_{k-1} \leq s, t \leq t_{k+2}\} \\ &\leq 2 \sup\{|B(s) - B(t_{k-1})| : t_{k-1} \leq s \leq t_{k+2}\} \end{aligned}$$

Noticing $t_{k+2} - t_{k-1} = \delta t_{k-1}$, where $\delta = (1 + \epsilon)^3 - 1$, scaling implies

$$\begin{aligned} &P \left(\max_{t_{k-1} \leq s \leq t_{k+2}} |B(s) - B(t)| > (3\delta t_{k-1} \log \log t_{k-1})^{1/2} \right) \\ &= P \left(\max_{0 \leq r \leq 1} |B(r)| > (3 \log \log t_{k-1})^{1/2} \right) \\ &\leq 2\kappa (3 \log \log t_{k-1})^{-1/2} \exp(-3 \log \log t_{k-1}/2) \end{aligned}$$

by a now familiar application of (8.5.1) and (8.5.2). Summing over k and using (b) gives

$$\limsup_{t \rightarrow \infty} (S_{[t]} - B_t)/(t \log \log t)^{1/2} \leq (3\delta)^{1/2}$$

If we recall $\delta = (1 + \epsilon)^3 - 1$ and let $\epsilon \downarrow 0$, (a) follows and the proof is complete. $\square$

Strassen (1965) has shown an exact converse. If Theorem 8.5.2 holds, then $EX_i = 0$ and $EX_i^2 = 1$. Another one of his contributions to this subject is:

Theorem 8.5.3 (Strassen's (1964) invariance principle) *Let $X_1, X_2, \dots$ be i.i.d. with $EX_i = 0$ and $EX_i^2 = 1$, let $S_n = X_1 + \dots + X_n$, and let $S_{(n\cdot)}$ be the usual linear interpolation. The limit set (i.e., the collection of limits of convergent subsequences) of*

$$Z_n(\cdot) = (2n \log \log n)^{-1/2} S_{(n\cdot)} \quad \text{for } n \geq 3$$

is $\mathcal{K} = \{f : f(x) = \int_0^x g(y) dy \text{ with } \int_0^1 g(y)^2 dy \leq 1\}$.

Jensen's inequality implies $f(1)^2 \leq \int_0^1 g(y)^2 dy \leq 1$ with equality if and only if $f(t) = t$, so Theorem 8.5.3 contains Theorem 8.5.2 as a special case and provides some information about how the large value of S_n came about.

Exercises

8.5.1 Show that if $E|X_i|^\alpha = \infty$ for some $\alpha < 2$, then

$$\limsup_{n \rightarrow \infty} |X_n|/n^{1/\alpha} = \infty \quad \text{a.s.}$$

so the law of the iterated logarithm fails.

8.5.2 Give a direct proof that, under the hypotheses of Theorem 8.5.3, the limit set of $\{S_n/(2n \log \log n)^{1/2}\}$ is $[-1, 1]$.